

Raunak Kumar

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<https://raunakmr.github.io>

SUMMARY I take keen interest in machine learning, algorithms and competitive programming. My goal is to contribute to the advancement in machine learning research in order to develop novel solutions to real-life problems.

EDUCATION *B.Sc. Computer Science* September 2013 - May 2018
University of British Columbia, Vancouver, BC

- Grade in Computer Science and Mathematics courses: 90.4%
- Selected coursework:
 - Graduate courses: Machine Learning, Randomized Big Data Algorithms, Probabilistic Graphical Models.
 - Advanced Algorithm Design and Analysis, Theory of Computation, Distributed Systems, Advanced Operating Systems.
 - Probability, Graph Theory, Real Analysis.

AWARDS (Nominated¹) Computing Research Association Outstanding Undergraduate Researcher Award (2018)
UBC Computer Science Teaching Assistant Award (2017)
NSERC Undergraduate Student Research Award (2017)
Science Scholar / Dean's Honors List (2013 - 2017)
J. Fred Muir Memorial Scholarship in Science (2014)
Trek Excellence Scholarship (2014) - Top 5% of undergraduates
UBC Chancellor's Scholar Award (2013)

PUBLICATIONS • **Raunak Kumar**, Mark Schmidt. *Convergence Rate of Expectation-Maximization*. NIPS Workshop on Optimization for Machine Learning. 2017.

RESEARCH EXPERIENCE *NSERC Research Assistantship* May 2017 - Present
Computer Science Department, UBC
Supervisor: Mark Schmidt

- I am currently working on developing an improved variant of the expectation-maximization (EM) algorithm using techniques from numerical optimization. I am also investigating if deep neural networks can learn to recognize characters from a video stream without supervision.
- We view EM in a generalized surrogate optimization framework and provide the first convergence rate for non-convex functions in this framework and, consequently, also for EM. We have published a paper titled *Convergence Rate of Expectation-Maximization* at a workshop for optimization in machine learning at Neural Information Processing Systems (NIPS) 2017.
- I collaborated with one of Prof. Schmidt's graduate students, Alireza Shafaei, on improving deep learning architectures for image segmentation. By incorporating the structure of the output space into the network, we obtained a 7%

¹decision expected to be announced in mid-December at <https://cra.org/about/awards/outstanding-undergraduate-researcher-award/>

increase in performance of ResNet-38 on the standard Cityscapes dataset. Unfortunately, we didn't observe the same improvement in PSPNet, the state of the art network for image segmentation.

- In July 2017, I consulted for a company in Victoria, BC on detecting dangerous gases in the air based on sensor measurements as part of a smart air purification system. I cleaned the provided data, produced visualizations using dimensionality reduction techniques like PCA and ISOMAP, and implemented random forest and a single layer neural network classification. We were able to achieve 99% classification accuracy on validation data. Furthermore, I advised the company that their data was biased, and that they should resort to an alternative data collection process and they are currently doing so.

INDUSTRY EXPERIENCE

Software Engineering Intern
Google, Mountain View, CA

May 2016 - August 2016

- I wrote a distributed datastore trigger utility in C++ for the App and Product Infrastructure team. The utility provides an extensible way to maintain triggers, which are used to schedule and perform asynchronous actions. The utility was used to remove 1 billion triggers from storage.

TEACHING

Instructor / Coordinator
Computer Science Department, UBC

January 2017 - April 2017

- I co-taught CPSC 490: Problem Solving in Computer Science, focusing on practical applications of advanced algorithms and data structures.
- My responsibilities included lecturing a class of 15-30 students 3 times a week, and writing lecture notes and assignments.
- Website: www.ugrad.cs.ubc.ca/~cs490/2016W2/.

Teaching Assistant
Computer Science Department, UBC

September 2014 - Present

- I have worked as a TA for 9 terms for the following courses: Advanced Algorithm Design and Analysis, Machine Learning, Advanced Operating Systems, Intermediate Algorithm Design and Analysis, Introduction to Computer Systems, and Models of Computation. Usually, only graduate students are assigned to TA the first 3 courses.
- In January 2018, I will be a TA for CPSC 540, a graduate course on machine learning even though I am only an undergraduate student.
- My responsibilities include lecturing sections ranging from 15 - 100 students, holding office hours and grading. I have taught an entire lecture for two algorithms courses as well.

PRESENTATIONS

- UBC Machine Learning Reading Group:
 - *Density Estimation using Real NVP*. November 2017.
 - *Adagrad², Adam, Online-to-Batch*. July 2017.
 - *Temporal-Difference Learning*. January 2017.
- UBC Machine Learning Theory Reading Group:
 - *VC-Dimension and the Fundamental Theorem of Learning*. January 2017.

²Found an error in the algorithm as presented in *Online Convex Optimization* by Elad Hazan.

PROJECTS

Programming Competitions

- UBC ACM programming team
 - Contestant: 2015 - Present.
 - Coach (Div. 2): 2017 - Present.
- ACM-ICPC PacNW Regionals
 - Contestant (Div. 1): 13th (2017), 15th (2016).
 - Contestant (Div. 2): 6th (2015).
 - Coach (Div. 2): 1st, 3rd (2017).
- Microsoft College Code Competition: 2nd (2017), 6th (2016).
- SFU Winter Programming Contest: 8th (2017).
- NAIPC (Open Division USA/Canada): 27th (2016).
- Calgary Collegiate Programming Competition: 3rd (2015, Div. 2).

Approximating Steiner Trees in the Semi-Streaming Model

- I presented a $2(1 + \epsilon) \left(2^{\frac{\log n}{\log \log n}} + 1 \right)$ -approximation for the weighted metric Steiner tree problem using $O(n \text{ polylog} n)$ space in the semi-streaming model, as part of a course project for a graduate class on randomized big data algorithms. The paper can be found on my website.

UBC Machine Learning Club

- I co-founded a machine learning club at UBC aimed at providing students an opportunity to gain experience with practical machine learning.
- Beginning November 2017, we aim to host semi-monthly challenges where students are required to submit code to solve machine learning tasks on common datasets.

Kernel

- I implemented basic functionality of a kernel using C and x86 assembly on the Bochs emulator. Features included memory management, process management, system calls, interrupt handling, signals and hardware interrupts.

Web Proxy Server

- I implemented a multi-threaded server application using C and Unix sockets to access HTTP content from websites, block blacklisted websites and cache results.

DNS Client

- I implemented a scaled down DNS resolver using Java and UDP datagrams to interact with various DNS servers to resolve domain names into IP addresses.

Dictionary Client

- I implemented a client side application using Java and TCP to connect with specified dictionary servers and retrieve word definitions.

insightUBC

- I implemented a backend web application using Typescript and Node.js to query UBC rooms and courses data.

TECHNOLOGY *Programming Languages:* C++, Python, C, Matlab.
SKILLS *Software:* PyTorch, Numpy, Scikit-Learn, TensorFlow, Vim, Git.